

## Number System

1. Find the number of divisors of 1728?
2. Find the sum of divisors of 544 excluding 1 and 544?
3. Find the sum of the sum of even divisors of 96 and the sum of odd divisors of 3600?
4. Find the maximum value of  $n$  such that  $157!$  is perfectly divisible by  $10^n$ ?
5. Find the maximum power of 18 in  $200!$ ?
6. Find the number of zeroes at the end of  $48!$ ?
7. The unit's digit of the expression  $127^{813} \times 553^{3703} \times 4532^{828}$  is?
8. A hundred and twenty-digit number is formed by writing the first  $x$  natural numbers in front of each other as 12345678910111213.... Find the remainder when this number is divided by 8?
9. If  $x = (16^3 + 17^3 + 18^3 + 19^3)$ , then  $x$  divided by 70 leaves a remainder of?
10. What is the remainder when  $(1!)^3 + (2!)^3 + (3!)^3 + (4!)^3 + \dots + (1152!)^3$  is divided by 1152?
11. The remainder when  $10^{10} + 10^{100} + 10^{1000} + \dots + 10^{10000000000}$  is divided by 7 is
12. After the division of a number successively by 3, 4 and 7, the remainders obtained are 2, 1 and 4 respectively. What will be the remainder if 84 divide the same number?

13. What are the last two digits of  $7^{2008}$ ?
14. A 3-digit number  $4a3$  is added to another 3-digit number 984 to give a 4-digit number  $13b7$ , which is divisible by 11. Then  $(a + b) = ?$
15. Which digit should come in the place of \* and # if the number  $62684* \#$  is divisible by both 8 and 5?
16. What is the least number which when doubled will exactly be divisible by 12, 14, 18 and 22?
17. The sides of a pentagonal field (not regular) are 1737 meters, 2160 metres, 2358 metres, 1422 metres and 2214 metres respectively. Find the greatest length of the tape by which the five sides may be measured completely.
18. If  $x$  is a number of five-digits, which when divided by 8, 12, 15 and 20 leaves respectively 5, 9, 12 and 17 as remainders, then find  $x$  such that it is the lowest such number?
19. Three numbers which are co-prime to each other are such that the product of the first two is 119 and that of the last two is 391. What is the sum of the three numbers?
20. A rectangular courtyard 3.78 meters long 5.25 meters wide is to be paved exactly with square tiles, all of the same size. what is the largest size of the tile which could be used for the purpose?

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Answer:

| Q-no | Answer | Q-No | Answer  |
|------|--------|------|---------|
| 1    | 28     | 11   | 5       |
| 2    | 589    | 12   | 53      |
| 3    | 651    | 13   | 01      |
| 4    | 38     | 14   | 10      |
| 5    | 48     | 15   | 4 and 0 |
| 6    | 10     | 16   | 1386    |
| 7    | 4      | 17   | 9       |
| 8    | 6      | 18   | 10077   |
| 9    | 0      | 19   | 47      |
| 10   | 225    | 20   | 21 cm   |